

Lab guide 4:

Services

Q.1 Create a deployment using image quay.io/mayank123modi/simple-webapp named test3 After creating this expose it on ClusterIP type service by CLI method and check everything .

Q.2 create a deployment using mdhack/myapache and expose it using NodePort type Service

Q.3 Expose a deployment that named test3 service name should be kubecluster and type is NodePort by YAML creation.

Ans: 1

kubectl create deployment --image quay.io/mayank123modi/simple-webapp test3

kubectl expose deployment test3 --port 8080

kubectl get svc

kubectl describe svc test3

kubectl get pods -o wide

curl 192.168.2.7:8080

```
root@master:~# kubectl create deployment quiz --image mdhack/myques:v1 -n mayank
deployment.apps/quiz created
```

```

root@master:~# kubectl expose deployment test3 --port 8080
service/test3 exposed
root@master:~# kubectl get svc
NAME      TYPE        CLUSTER-IP      EXTERNAL-IP  PORT(S)    AGE
test3     ClusterIP   10.111.166.170   <none>       8080/TCP   3s
root@master:~# kubectl describe svc test3
Name:      test3
Namespace: learning
Labels:    app=test3
Annotations: <none>
Selector:  app=test3
Type:      ClusterIP
IP Family Policy: SingleStack
IP Families: IPv4
IP:        10.111.166.170
IPs:       10.111.166.170
Port:      <unset> 8080/TCP
TargetPort: 8080/TCP
Endpoints: 192.168.2.7:8080
Session Affinity: None
Events:    <none>
root@master:~# kubectl get pods -o wide
NAME                                READY   STATUS    RESTARTS   AGE   IP            NODE     NOMINATED NODE   READINESS GATES
test3-77dcf676c9-pmdqx             1/1     Running   0           88s   192.168.2.7   node2    <none>            <none>

```

```

root@master:~# curl 192.168.2.7:8080
<!doctype html>
<title>Hello from Flask</title>
<body style="background: #130f40;"></body>
<div style="color: #e4e4e4;
  text-align: center;
  height: 90px;
  vertical-align: middle;">

  <h1>Hello from test3-77dcf676c9-pmdqx!</h1>

```

Ans: 2

kubectl create deployment --image mdhack/myserver test4

kubectl expose deployment test4 --port 80 --type NodePort

kubectl get svc

kubectl describe svc test4

kubectl get pods -o wide

curl <your pod ip address>

```

root@master:~# kubectl create deployment --image mdhack/myserver test4
deployment.apps/test4 created
root@master:~# kubectl expose deployment test4 --port 80
service/test4 exposed
root@master:~# kubectl get svc
NAME      TYPE        CLUSTER-IP      EXTERNAL-IP  PORT(S)    AGE
test3     ClusterIP   10.111.166.170   <none>       8080/TCP   3m20s
test4     ClusterIP   10.101.157.76    <none>       80/TCP     2s
root@master:~# kubectl describe svc test4
Name:      test4
Namespace: learning
Labels:    app=test4
Annotations: <none>
Selector:  app=test4
Type:      ClusterIP
IP Family Policy: SingleStack
IP Families: IPv4
IP:         10.101.157.76
IPs:        10.101.157.76
Port:      <unset> 80/TCP
TargetPort: 80/TCP
Endpoints: 192.168.2.8:80
Session Affinity: None
Events:    <none>
root@master:~# kubectl get pods -o wide
NAME      READY   STATUS    RESTARTS   AGE   IP            NODE   NOMINATED NODE   READINESS GATES
test3-77dcf676c9-pmdqx  1/1     Running   0           4m45s  192.168.2.7   node2   <none>            <none>
test4-6886d5645-wk6c2   1/1     Running   0           19s    192.168.2.8   node2   <none>            <none>

```

```

root@master:~# curl 192.168.2.8
<body bgcolor='aqua'>
<pre>

Welcome to MDhack/Mayank web server for testing</br>
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 9001
    inet 192.168.2.8 netmask 255.255.255.255 broadcast 192.168.2.8
    inet6 fe80::4464:8bff:fed6:d925 prefixlen 64 scopeid 0x20<link>
    ether 46:64:8b:d6:d9:25 txqueuelen 0 (Ethernet)
    RX packets 10 bytes 811 (811.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 13 bytes 950 (950.0 B)
    TX errors 0 dropped 1 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

</br>
You will Definately enjoy this training

</pre>

```

Ans: 3

kubectl create service nodeport kubeccluster --tcp 8080 --dry-run -o yaml > svc.yaml

vim svc.yaml

kubectl create -f svc.yaml

kubectl get svc

kubectl describe svc kubeccluster

kubectl get pods -o wide

curl <publicip of cluster>:<nodeport>

```
root@master:~# kubectl create service nodeport kubeccluster --tcp 8080 --dry-run -o yaml > svc.yaml
W0505 05:36:17.348050 11291 helpers.go:636] --dry-run is deprecated and can be replaced with --dry-run=client.
root@master:~# vim svc.yaml
```

```
root@master:~# kubectl delete svc kubeccluster
service "kubeccluster" deleted
root@master:~# kubectl create -f svc.yaml
service/kubeccluster created
root@master:~# kubectl get svc
NAME         TYPE        CLUSTER-IP      EXTERNAL-IP  PORT(S)          AGE
kubeccluster NodePort    10.107.17.127   <none>       8080:30978/TCP   3s
test3        ClusterIP   10.111.166.170  <none>       8080/TCP         8m53s
test4        ClusterIP   10.101.157.76   <none>       80/TCP           5m35s
root@master:~# kubectl describe svc kubeccluster
Name:         kubeccluster
Namespace:    learning
Labels:       app=kubeccluster
Annotations:  <none>
Selector:     app=test3
Type:         NodePort
IP Family Policy: SingleStack
IP Families:  IPv4
IP:           10.107.17.127
IPs:          10.107.17.127
Port:         8080 8080/TCP
TargetPort:   8080/TCP
NodePort:     8080 30978/TCP
Endpoints:    192.168.2.7:8080
Session Affinity: None
External Traffic Policy: Cluster
Events:       <none>
root@master:~# kubectl get pods -o wide
NAME                                READY   STATUS    RESTARTS   AGE   IP            NODE     NOMINATED NODE   READINESS GATES
test3-77dcf676c9-pmdqx              1/1    Running   0          10m   192.168.2.7   node2    <none>            <none>
test4-6886d5645-wk6c2              1/1    Running   0          5m58s 192.168.2.8   node2    <none>            <none>
```

```
root@master:~# curl 3.143.12.125:30978
<!doctype html>
<title>Hello from Flask</title>
<body style="background: #130f40;"></body>
<div style="color: #e4e4e4;
  text-align: center;
  height: 90px;
  vertical-align: middle;">

  <h1>Hello from test3-77dcf676c9-pmdqx!</h1>
```

```
root@master:~# cat svc.yaml
apiVersion: v1
kind: Service
metadata:
  creationTimestamp: null
  labels:
    app: kubeccluster
  name: kubeccluster
spec:
  ports:
    - name: "8080"
      port: 8080
      protocol: TCP
      targetPort: 8080
  selector:
    app: test3 #same labels as your deployment have
  type: NodePort
status:
  loadBalancer: {}
root@master:~# █
```

Clean up your environment

kubectl delete all --all --force

```
root@master:~# kubectl delete all --all --force
warning: Immediate deletion does not wait for confirmation that the running resource
n on the cluster indefinitely.
pod "test3-77dcf676c9-pmdqx" force deleted
pod "test4-6886d5645-wk6c2" force deleted
service "kubecuster" force deleted
service "test3" force deleted
service "test4" force deleted
deployment.apps "test3" force deleted
deployment.apps "test4" force deleted
root@master:~#
```